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PALAEOMAGNETISM OF ORDOVICIAN IGNIMBRITES FROM KILLARY HARBOUR, EIRE

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Twenty-five samples have been collected from four bands of highly oxidized mid-Ordovician (post-Arenig) ignimbrite exposed in County Mayo, Eire. The natural remanent magnetization (NRM) of three of the bands was significantly misaligned with the present Earth's field. Sixty-nine specimens were then subjected to detailed demagnetization in alternating fields to 795 peak Oe and temperatures to 670°C. The results demonstrated that the NRM of the four bands is dominated by a very stable component with blocking temperature in the 400–550°C range. A positive fold test indicates that this magnetization was acquired prior to the folding of the Mweelrea syncline in the Ordovician or earliest Silurian. Assuming a geocentric axial dipole field of "reversed" polarity and allowing for possible incomplete averaging of the secular variation or other field assymmetries, a geographic north pole in the Pacific Ocean (9°N, 146°W, dp = 12°, dm = 22°) was calculated from the resultant of the four band mean directions relative to bedding planes, after treatment of all samples to 540 Oe. This result is compatible with the few published Ordovician poles relative to Europe.

1. Introduction

Palaeomagnetic data from Ireland are desirable because of that island's critical location in any palaeogeography of the North Atlantic. However, apart from detailed studies of Tertiary rocks in Northern Ireland [1, 2], no Irish results appear to have been published. Although Tertiary data are relevant if one attempts to establish a chronology of the opening of the North Atlantic, the detection of possible earlier continental drift episodes requires evidence from Palaeozoic and older rocks.

Unfortunately, Lower Palaeozoic rock magnetism has not been extensively studied, and existing results are largely based upon preliminary work with sedimentary rocks. For the Ordovician relative to northern Eurasia the literature includes about one dozen pole positions [3–8] meeting minimum-reliability criteria [3]. Among these, one British study [4] is based on volcanic rocks. More Lower Palaeozoic results from volcanic rocks would be helpful, especially since lavas of certain compositions are among the rock types most likely to reveal a stable, primary magnetization.

In 1965 some Ordovician ignimbrites exposed near Killary Harbour in County Mayo, Eire, were samples by one of us (E.R.D.), with the kind collaboration of Professor Marshall Kay of Columbia University, New York. Previous investigations of (Pleistocene) ignimbrites [9, 10] have shown that such rocks can be capable of retaining a stable magnetization, at least since the Pleistocene. An interesting aspect of the present collection is that it enables us to assess the palaeomagnetic usefulness of a much older ignimbrite body. In this paper the results of magnetic studies of the Killary Harbour rocks are discussed.

2. Geology and sampling

The Lower Palaeozoic stratigraphy near Killary Harbour has been described by Dewey [11]. Six ignimbrite bands interbedded with grits of the Mweelrea Group (fig. 1) are exposed on both limbs of

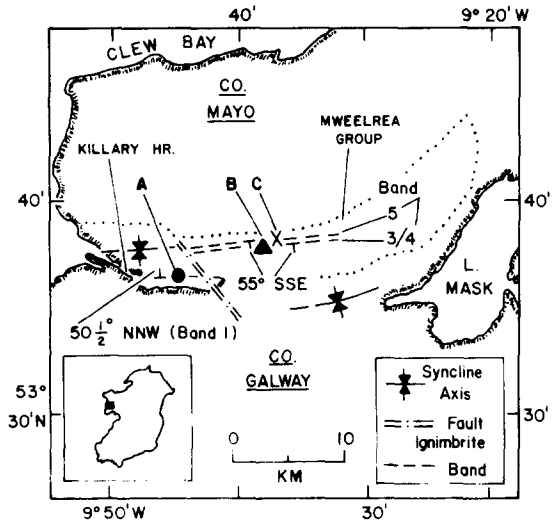


Fig. 1. Location of Ordovician ignimbrites samples at sites A, B,C, near Killary Harbour, Eire. Map modified after Dewey [11].

the Mweelrea syncline, whose axis runs west-east for many kilometers with nearly zero plunge. The Ordovician age of the Mweelrea grits has been established by fossil evidence, but their position within that period is uncertain. It could be Caradocian [11], but Professor Alwyn Williams of Queen's University, Belfast (personal communication), suggests that at least part of the Mweelrea succession may be older, namely Llanvirnian. Stanton [12] dates the grits as Middle Ordovician, along with the underlying Glenummera Series. The pronounced Ordovician-Silurian unconformity in the area indicates that much of the folding of the Mweelrea syncline occurred before the Upper Llandovery (Lower Silurian) [11]. Hence it will be assumed that the ignimbrites originated in post-Arenig, mid-Ordovician time and were first deformed in the Ordovician or earliest Silurian.

Twenty-five hand samples were obtained at three sites (fig. 1). The lowest ignimbrite band (no. 1) is from site A on the southern limb of the Mweelrea syncline. Sites B (bands 3, 4?) and C (band 5), on the northern limb, are about 200 m apart and nearly 10 km east of site A. Bands 3 and 4 are not necessarily distinct on the northern limb, where Stanton [12] suggests that band 4 is absent. However, Dewey [11] finds that Stanton's "band 3" includes an upper "unit" (band 4?) lying above a weathering zone that may represent a "considerable" time interval. Pending a test (section 3.4) the two units were sampled separately at site B.

The samples were oriented *in situ* by a Brunton compass. Repeated comparisons of direct compass readings with those "sighted in" from some distance beyond the outcrop failed to detect significant (> 1°) compass deflection errors due to the rocks. Strike and dip readings gave 263°, 50½°NNW at site A, and 75°, 55°SSE at sites B and C.

3. AF demagnetization

3.1. *Natural remanent magnetization (NRM)*

At the laboratory the samples were cut into cylindrical specimens of 2.2 cm diameter and 1.8 cm mean height. Fifty specimens (two per sample) were used for alternating-field (AF) demagnetization. No results were discarded. An astatic magnetometer with transverse system [13] rendered directions of magnetization to 1° and intensities to ± 3% over the observed NRM ranges, which were $J_N = 3 \times 10^{-5}$–24×10^{-5} emu cm^{-3} (sites A and B) and $J_N = 1 \times 10^{-5}$–7×10^{-5} emu cm^{-3} (site C). For each sample a direction was obtained by averaging the two specimen directions.

Site and band mean directions of NRM relative to bedding planes (table 1) were found from the *in situ* directions after a simple rotation about strike, justified by the near-horizontality of the fold axis. For bands 3–5 the sample directions are well-grouped, corresponding to fairly large values of the precision parameter, k [14]. The alignment of the four band mean directions is also quite close ($k = 50$), whereas that for band 1 is extremely poor ($k = 2$).

Site mean directions of NRM relative to the *present* horizontal are numbered 0 in fig. 3a. The site A direction makes large angles with both the present field and a geocentric axial dipole field, indicating partial stability despite the large directional scatter. A smaller but still significant discrepancy exists between these fields and the site B direction ($\alpha_{95} = 5°$; table 1). No such misalignment is found at site C, and here one must suspect a relatively recent origin of the NRM. To test this possibility and remove unstable components generally, the ignimbrites were subjected to AF treatment.

Table 1
Site and band mean directions of magnetization and statistical parameters [14].

Site No.	Band No.	N	NRM					After 540 Oe (peak) treatment				
			D	I	R	k	α_{95}	D	I	R	k	α_{95}
A	1	9	129	+34	5.330	2	47°	128	+35	8.074	9	19°
B	3	4	138	+27	3.977	130	8°	141	+15	3.958	72	11°
B	4	4	134	+32	3.982	163	7°	128	+27	3.975	120	8°
B	3 + 4	8	136	+29	7.947	131	5°	135	+22	7.843	45	8°
C	5	8	157	+42	7.871	54	8°	144	+52	7.667	21	12°
All samples		25	141	+36	20.775	6	13°	135	+36	22.949	12	9°
All bands		4	138	+34	3.940	50	13°	134	+32	3.870	23	20°
			NRM					After 450°C treatment				
All samples		11	148	+34	10.644	28	9°	141	+33	10.324	15	12°

N = number of samples or bands averaged before and after AF demagnetization (all sample directions based on 2 specimens/sample) or thermal demagnetization (1–2 specimens/sample); D and I (in degrees) = declination (E of N) and inclination (positive downward) of the resultant of N unit remanence vectors relative to bedding planes; R = length of vector resultant; k = precision parameter; α_{95} = radius of 95% confidence circle.

3.2. *AF pilot study*

Five specimens were demagnetized stepwise to nearly 800 peak Oe, using standard AF techniques with a 3-axis tumbler unit. The normalized intensity curves (fig. 2a) of specimens 2a (site A) and 25 a (site B) are similar, J/J_N dropping to 0.5 at about 400 Oe. Specimen 16 b (site C) retains 60% of its NRM intensity at 795 Oe, indicating a particularly hard magnetization. These trends, along with the small range of directional changes (fig. 2b), reflect the dominance of very stable components in the remanence.

The two remaining specimens (not shown) from sites A and C, respectively, were found to contain prominent low-coercivity components directed close to the present Earth's field. These were removed by AF treatment to 150 Oe, leaving a stable magnetization. Between 25 and 150 Oe, the steeply inclined NRM of the site A specimen changed continuously into a vector of low inclination, then maintained up to 795 Oe a direction similar to that of specimen 2a (fig. 2b), from the same site. A plausible explanation for the low site A precision of the NRM is that steeply inclined viscous components are superimposed upon the gently inclined stable remanence of the rocks.

3.3. *General AF treatment*

The largely stable magnetic behaviour of the pilot specimens made it difficult to determine an optimum demagnetizing field, though this was estimated to be in the 500–550 Oe range. Accordingly, all 50 specimens were demagnetized to 310, 540 and 770 Oe, the first and third steps being for comparison. All three site mean directions show little change up to the highest field (fig. 3a), but with varying precision. Values of k for site A, B and C mean directions after treatment to 310 Oe (770 Oe) were 7, 41 and 21 (3, 10 and 16), respectively. Comparison with table 1 shows the 540 Oe treatment to have optimized the site A grouping (k = 9), whereas the NRM has greatest precision in the case of sites B(131) and C(54). Nevertheless, the data from the 540 Oe step were adopted for all specimens, as the results of neither magnetic nor thermal pilot studies (fig. 2) seem to justify treating different ignimbrite bands differently. Moreover, since the site B and C *in situ* directions are steep, their reduced precision after AF treatment may in fact be compatible with *improved* stability due to the removal of viscous components aligned with a recent Earth's field.

The 25 sample mean directions relative to bedding

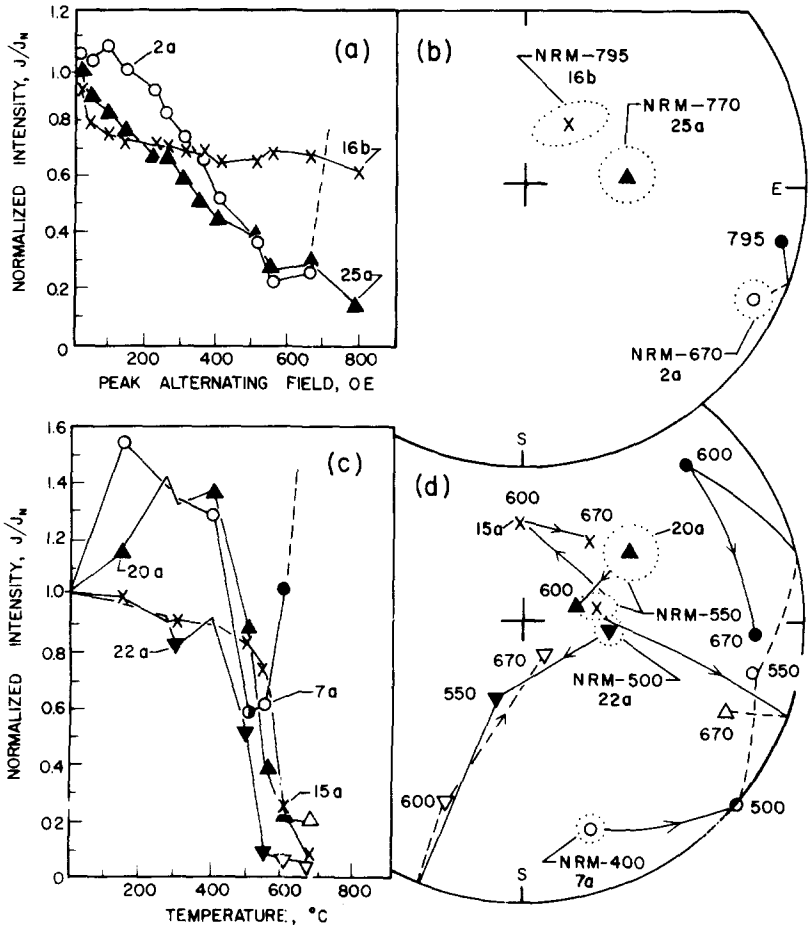


Fig. 2. Behaviour of pilot specimens during stepwise demagnetization. (a) Normalized intensity and (b) direction of magnetization for three specimens after AC treatment. X and solid symbols indicate N pole is downward; open symbols, upward. Numbers are in peak Oe. (c)(d) Corresponding magnetization changes for four specimens after thermal treatment, with numbers in °C. (b) and (d) are equal-area plots with the present horizontal as circumference. Successive steps have been connected by great-circle routes. Within dotted outlines, directions remain stable up to the highest step shown. ●○ site A, band 1; ▼▽ site B, band 3; ▲△ site B, band 4; X site C, band 5.

are spread out over the second quadrant (fig. 3c), with mean direction $D = 135°, I = +36°$ and $k = 12$ (table 1). The more realistic procedure of assigning unit weight to bands ($N = 4$) rather than samples gives a similar direction ($D = 134°, I = +32°$), but with closer vector grouping ($k = 23$).

3.4. *Separation of bands 3 and 4*

So far it has been assumed that ignimbrite bands 3 and 4 are distinct at site B. In that case, the secular variation may have produced a significant difference between the two mean directions of magnetization. To examine this statistically, four samples each from Dewey's "lower unit" (band 3) and "upper unit" (band 3 or 4) were subjected to a variance-ratio ("*F*"-ratio) test [15, 16].

To apply this test one must assume that the variances of the two populations being compared are approximately equal. By applying initial *F*-ratio tests to both NRM and AF-treated sample directions

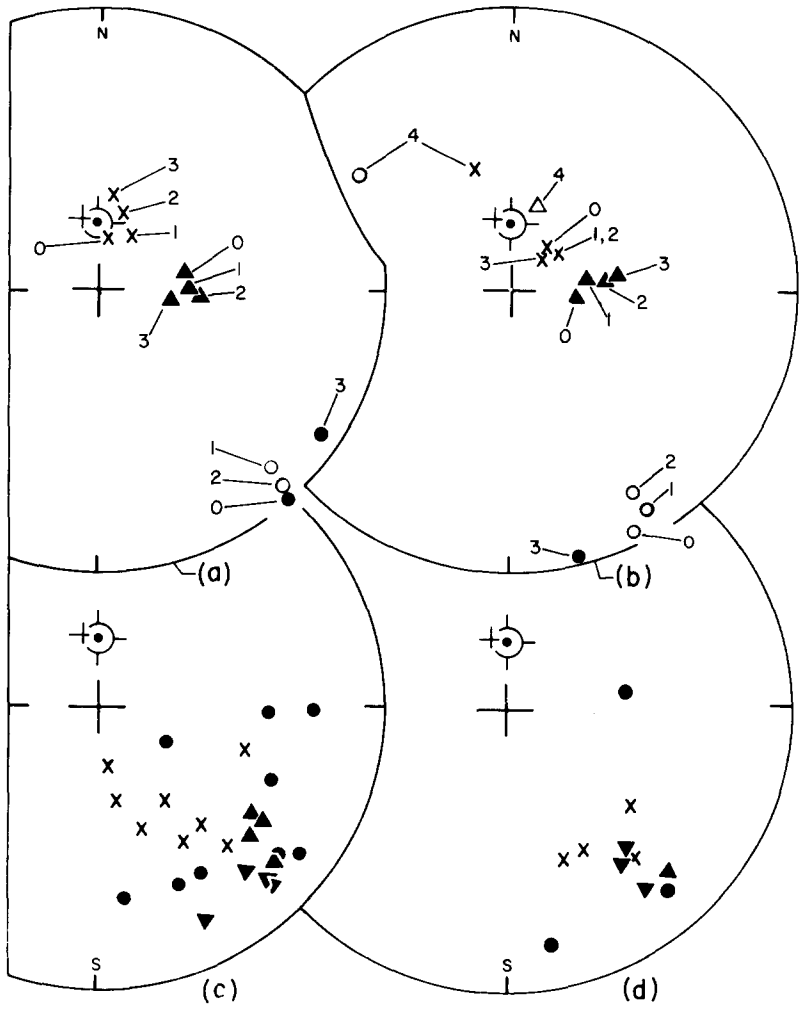


Fig. 3. (a) (b) Equal-area plots of mean site directions relative to the present horizontal, for NRM (step 0) and (a) after AC treatment at 310, 540, and 770 peak Oe (steps 1–3); (b) after thermal treatment at 400, 450, 500 and 670° (steps 1–4). (c) (d) Mean sample directions relative to bedding planes, for the samples averaged in (a) and (b), respectively, after demagnetization at (c) 540 Oe; (d) 450°C. Band and polarity symbols are as in fig. 2, except that in figs. 3a and 3b, bands 3 and 4 have been combined into a single site B symbol (upright triangles). ⊕, geocentric axial dipole field; +, 1965 field.

for the two bands, this assumption was shown to be valid at the 99% probability level. One may define "between-band" and "within-band" angular variances by [15, 16]

$$\bar{\delta}^2_b = 2\,(\Sigma R_i - R)\,/\,(B-1)$$

and

$$\bar{\delta}^2_w = 2\,(N - \Sigma R_i)\,/\,(N-B)$$

where R_i and R are the magnitudes of the unit vector resultants for the ith band and the N (= 8) samples, respectively, and B is the number of bands (= 2). Then if $\bar{\delta}^2_b/\bar{\delta}^2_w$ exceed the statistic $F_{2(B-1),(N-B),\alpha}$ (see standard tables), the B vectors may be said to differ significantly with probability greater than $1-\alpha$.

Table 2
F-ratio test at site B.

Bands	Treatment	B	N	δ_b^2	δ_w^2	δ_b^2/δ_w^2	$F_{2,12,0.01}$	$F_{2,12,0.05}$
3,4	NRM	2	8	0.0242	0.0138	1.75	6.93	3.88
3,4	540 Oe	2	8	0.1806	0.0222	8.13	6.93	3.88

B and N = number of bands and samples; δ_b^2 and δ_w^2 (in radians^2) = between-site and within-site angular variance; last two columns = F-ratio, $F2(B-1), 2(N-B), \alpha$ at 1% and 5% probability levels.

Table 2 shows that the two band NRM directions are not significantly distinct at the 5% probability level, whereas after 540 Oe treatment it becomes more than 99% probable that they *are* distinct. Therefore bands 3 and 4 will be assumed to be separately exposed at site B. However, this conclusion, being based on limited sampling at one site, does not necessarily apply to other exposures of these bands.

4. Origin of the magnetization

The ignimbrites have been described [11, 12] as welded tuffs resulting from a *nuée ardente*-type magma highly charged with gases. Dewey [11] observed magnetite concentrations around the margins of gas-explosion nuclei. Covering an area of some 200–300 km^2 without marked compositional variation, these tuffs appear to have been rapidly chilled, perhaps due to deposition in shallow water [12]. Such a mode of formation should lead to a stable thermoremance (TRM), residing in the small ferrimagnetic grains. Larson et al. [17] quote a large value of remanent coercivity (480 Oe) for a Cenozoic welded tuff sample in which more than half the grain diameters were $< 1\ \mu$.

Polished sections of one specimen each from sites A and B were kindly examined by Dr. Norman Watkins of Florida State University, revealing the presence of much hematite in relic structures or intergrown with original titanomagnetite, and incipient development of pseudobrookite. This corresponds to the maximum oxidation state of titanomagnetite ("Index VI") in Watkins and Haggerty's [18] classification. Normally such oxidation probably occurs during the initial cooling of a lava. Recent studies, mostly of basic lavas [17, 19], showed that there tends to be a strong positive correlation between an advanced high-temperature oxidation state and high magnetic stability.

Such a correlation is consistent also with the present findings involving relatively acidic rocks, though conceivably the ignimbrites might have been oxidized through reheating a geologically long time after their formation. We consider this to be unlikely, since the simple structure and relative freshness of the Mweelrea rocks point to the absence of serious post-Silurian tectonic activity.

4.1. *Thermal demagnetization*

As a further stability test, the now standard stepwise thermal technique was applied to previously untreated specimens. The remanence is measured before and after heating to successively higher temperatures and cooling in zero field. A scaled-down version (inner diameter 13.0 cm) of the non-inductive furnace described by Irving et al. [20] was built. The field was nulled by three helmholtz coil pairs averaging 1 m diameter, and was monitored during demagnetizations with a set of fluxgate detectors mounted outside the heating jacket.

Great care was taken to optimize field-nulling in the magnetically noisy Chemistry-Physics building. First a relatively gradient-free site for the furnace was located by a fluxgate magnetometer survey. Stray fields of several hundred gammas ($1\gamma = 10^{-5}$ Oe), originating mainly in the nichrome heating wire, were greatly decreased by thermally demagnetizing the empty furnace before each run. Further precautions (e.g. confining runs to quiet periods) reduced the maximum estimated field under working conditions to about 50γ, i.e. 0.1% of the laboratory field. This was acceptable, since exhaustive calibrations had indicated that $400-500\gamma$ fields would be needed even near blocking temperature to deflect the remanence of the ignimbrites significantly. Systematic directional

errors in any case would be minimized, as the remanence vectors were always randomly positioned in the furnace before treatment. Between runs, the specimens were stored in magnetically shielded cans in which the field is $< 10\gamma$.

All demagnetizations were carried out in air, as this seemed to be a safe procedure [21]. All four normalized intensity curves in fig. 2c (one specimen per band) drop sharply in the 400–600°C region; this and the tight directional grouping up to 400 to 550°C (fig. 2d) indicates that the principal carrier of the NRM is a hard, high-blocking-temperature component residing in titanomagnetite. Birina et al. [10] inferred minimum temperatures of 400–500°C for acquisition of the stable TRM in Pleistocene ignimbrites from Kamchatka.

Above 600°C the directions diverge, but three of the J/J_N curves have a "tail", perhaps due to hematite. The initial intensity increase for specimens 7a and 20a might be due to low-field removal of an oppositely directed viscous component. All four curves suggest the presence of a minor component blocked near 300°C. The sharp intensity rise of specimen 7a above 500°C is puzzling; in a redbed specimen, Opdyke [22] observed a similar effect, which he believed to be due to an unknown chemical change.

Fifteen additional specimens from 11 samples were demagnetized in four steps. The *in situ* site mean directions up to 500°C (numbered 1–3; fig. 3b) are stable as in the AF treatment (fig. 3a), but become scattered after step 4 (670°C). After step 2 (450°C) the sample directions relative to bedding (fig. 3d) are quite closely grouped except for two site A samples. The resultant direction (141°, +33°) is comparable to the other mean directions in table 1.

4.2. *Fold test*

Giving bands unit weight ($N = 4$), the precision of the mean *in situ* remanence direction for Killary Harbour after AF-treatment to 540 Oe is $k = 3.5$. The corresponding precision after correction for geological tilt (i.e. relative to bedding) is $k = 23$ (table 1). McElhinny's [23] significance criterion for the fold test [24] is here inapplicable, as the *in situ* vector distribution is not Fisherian. However, the tilt correction has obviously much improved the alignment of the mean vectors for bands 3–5 on the northern limb of the Mweelrea syncline with band 1 on the southern limb (compare also figs. 3a and c, or 3b and d; though the comparisons, being between site and sample averages, are inexact). Hence, clearly the tilt correction constitutes a positive fold test. We conclude, therefore, that the magnetization was acquired prior to folding in the Ordovician or earliest Silurian.

5. The Ordovician field

Since different bands were compared across the syncline, the above conclusion implies that the *in situ* vector misalignment caused by the folding significantly exceeds that due to the secular variation ($\sim 10^3$ years), or possibly due to field asymmetries with a longer time scale (10^5–10^6 years) [25]. This would mean that during the Ordovician the amplitude of these asymmetries was not substantially greater than in recent times. A related basic assumption is that the field was a geocentric axial dipole when averaged over 10^3–10^6 years. These assumptions are plausible, though for Lower Palaeozoic times it is difficult to test them palaeomagnetically; however, a dipolar late Cambrian to Ordovician field has been inferred from Siberian results [8].

Because of the interbedded sediments, the time interval separating ignimbrite bands 1 and 5 is probably ample for the average field to have been that of an axial dipole. Still, only four bands are involved, and the asymmetries may not have been fully averaged. With this reservation in mind, pole positions were calculated from the mean north direction (134°, +32°; table 1) based on the four band vectors after 540 Oe treatment:

pole = 9°S, 34°E;
antipole = 9°N, 146°W;
dp = 12°, dm = 22°,

where dp and dm are semi-axes of the 95% confidence oval. Assuming an axial dipole field of "reversed" polarity, the antipole corresponds to a geographical north pole in the present Pacific Ocean (fig. 4, No. 2).

Despite the limited sampling, we regard this pole position as dependable within the confidence oval because of the good evidence for stability. This oval is rather large, but so is the separation of the Irish pole

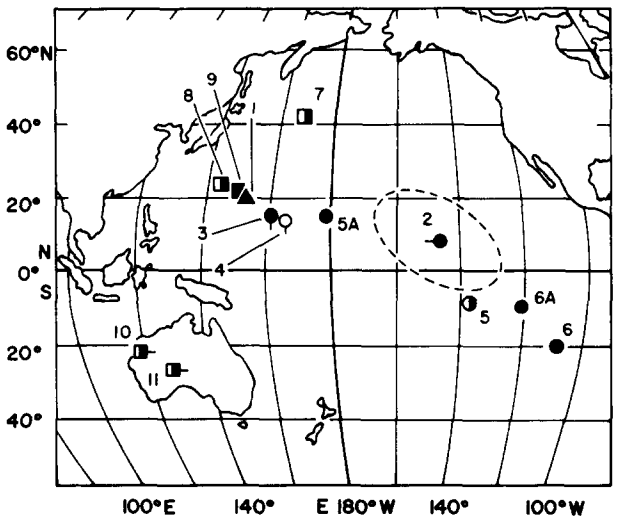


Fig. 4. Pole positions from Ordovician rocks in North America and northern Eurasia. Irving's [3] pole numbers are quoted for some earlier studies. 1. Juniata formation, Pennsylvania, pole 3.06 [3]; 2. Killary Harbour ignimbrites (with 95% confidence oval, this paper); 3. Borrowdale volcanics, NW England [4]; 4. Builth volcanics, Wales [4]; 5. Central Bohemian sediments, Lower Ordovician [5]; 5A. Oolitic ores, Bohemia [6]; 6. Central Bohemian sediments, Upper Ordovician [5]; 6A. Zahoramy layers, Bohemia [6]; 7. Tremadocian sediments, Baltic USSR, pole 3.01 [3]; 8. Asha stage, South Urals, pole 3.04 [3]; 9. South Urals and Baltic sediments [7]; 10. Lugov suite, Siberia, pole 3.10 [3]; 11. Sediments from Siberian platform [7, 8]. Poles shown by open (solid, half-open) symbols were inferred from rocks with presumed "normal" ("reversed", "mixed") polarity relative to the present field. In computing pole 2, unit weight was given to the four band mean directions (table 1).

from two British mid-Ordovician poles (fig. 4, Nos. 3 and 4) and from three of the poles for Czechoslovakia (fig. 4, Nos. 5, 6 and 6A). Whereas it is possible to devise palaeomagnetic schemes for testing certain models of "pre-Wegenerian" drift [26], the data in fig. 4 (including Killary Harbour) are still quite inadequate for such a test.

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